

Innovation

BACKTRACK: Your GPS for Learning
University of the Philippines Manila
202 words · Maximum 300 words

BACKTRACK's organizing question is: what is blocking the thing this learner needs now, and which useful step could get them back? The destination remains visible while the route changes with their answers.

Adaptive learning, prerequisite graphs, and goal-based tutoring already exist. ALEKS and modern AI learning tools overlap with this idea. We do not claim to have invented personalization. Our proposed contribution is the complete recovery workflow around Khan: a teacher-confirmed destination, an inspectable route, relevant practice, fresh evidence of return, and a reusable school kit.

The reward is concrete. When a learner demonstrates a step, its review leaves the route. When a learner needs help, a deeper step can appear. Confidence changes support and follow-up checks, not mathematical correctness. The current product requires two distinct unassisted checks for a route-specific decision with later checks to see what stays with the learner.

Learners choose a manageable block, can stop without losing progress, and may take one optional next turn. A comeback check preserves history while reassessing what is useful now. We will measure voluntary continuation, frustration, and return visits.

The working website lets judges experience those decisions. The pilot will test whether this combination adds value beyond an equal-time teacher-selected Khan routine.